



ENGINEERING MATHEMATICS - II

UE20MA151

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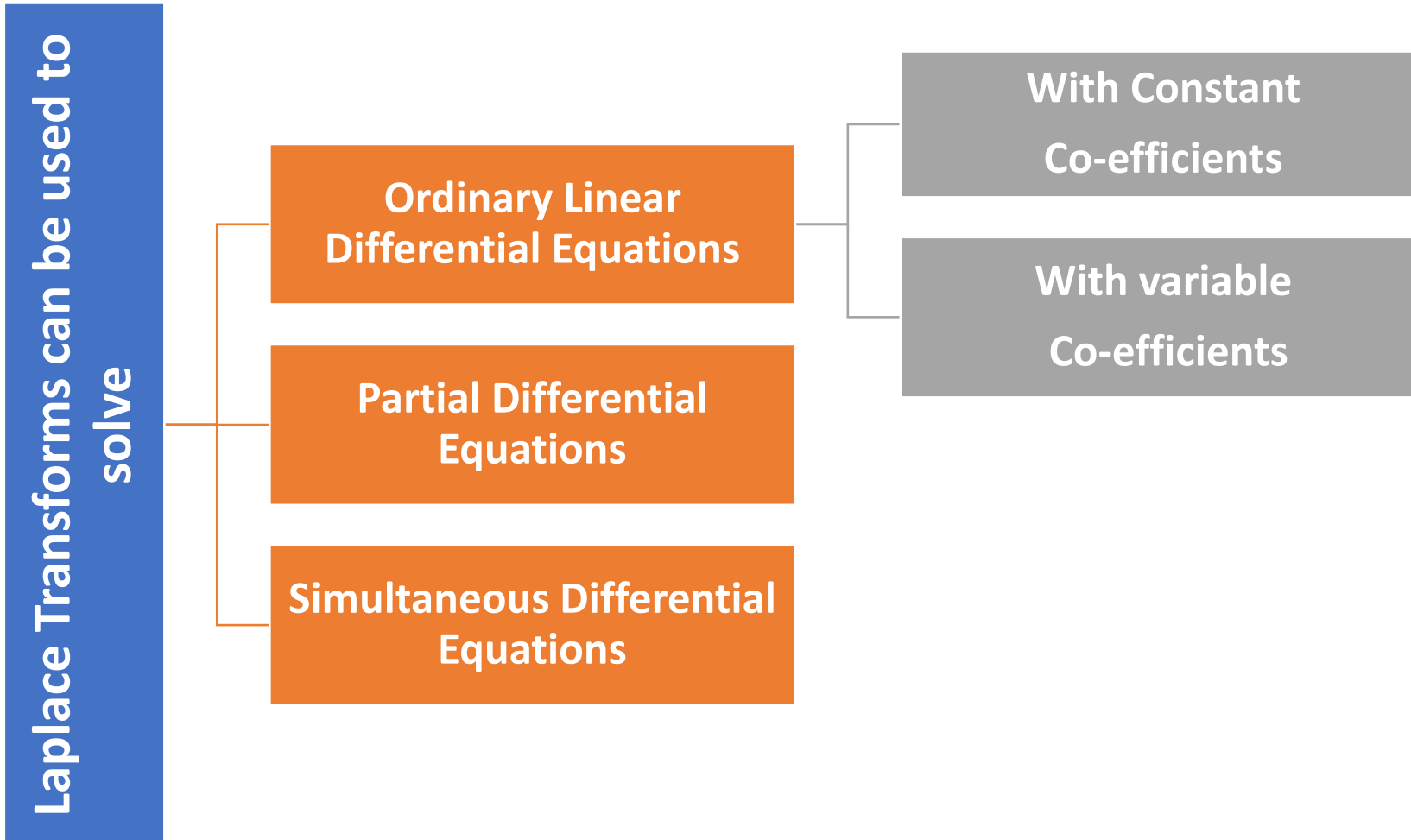
Unit 4 : Inverse Laplace Transform

Session : 9

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Application To Differential Equations



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INVERSE LAPLACE TRANSFORM

Time Domain

$$y^{(2)}(t) + y^{(1)}(t) + y(t) = x(t)$$

$$x(t) = 1$$

$$y(t) = \frac{1}{2} - e^{-t} + \frac{1}{2}e^{-2t}$$

Frequency Domain

$$H(s) = \frac{1}{s^2 + 3s + 2}$$

$$X(t) = \frac{1}{s}$$

Solve algebraic equation

$$\frac{1}{s} - \frac{1}{s^2 + 3s + 2}$$

Inverse Laplace transform

Laplace transform

Laplace transform of derivatives:

$$L\{y'(t)\} = sL[y(t)] - y(0)$$

$$L\{y''(t)\} = s^2L[y(t)] - sy(0) - y'(0)$$

$$L\{y'''(t)\} = s^3L[y(t)] - s^2y(0) - sy'(0) - y''(0)$$

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INVERSE LAPLACE TRANSFORM

Solution of Differential Equations Using Laplace Transforms- Working Procedure:

Step 1

- Given Differential Equation is expressed in the notation $y'(t), y''(t), \dots$ for the derivatives

Step 2

- Take Laplace Transforms on both sides. We use the expression for $L\{y'(t)\}$, $L\{y''(t)\}$...

Step 3

- Substitute the given initial conditions and simplify to obtain $L\{y(t)\}$ as a function of s .

Step 4

- Take inverse Laplace transform on both sides to obtain y .

1) Using Laplace Transforms

Solve $y'' - 2y' - 8y = 0, y(0) = 3, y'(0) = 6$.

Solution :

Taking Laplace Transforms of both sides,

$$L[y''(t)] - 2L[y'(t)] - 8L[y(t)] = 0$$

$$\blacktriangleright \{s^2 L[y(t)] - sy(0) - y'(0)\} - 2\{sL[y(t)] - y(0)\} - 8L[y(t)] = 0$$

$$\blacktriangleright s^2 L[y(t)] - 3s - 6\} - 2\{sL[y(t)] - 3\} - 8L[y(t)] = 0$$

$$\blacktriangleright (s^2 - 2s - 8)L[y(t)] - 3s = 0$$

$$\triangleright \{L[y(t)] = \frac{3s}{(s^2-2s-8)} = \frac{3s}{(s-4)(s+2)} = \frac{2}{s-4} + \frac{1}{s+2}$$

Take inverse Laplace transform on both sides ,

$$y(t) = 2L^{-1}\left(\frac{1}{s-4}\right) + L^{-1}\left(\frac{1}{s+2}\right)$$

$$y(t) = 2e^{4t} + e^{-2t}$$

2) Using Laplace Transforms,

Solve $y'' + 9y = \cos 2t$, if $y(0) = 1, y(\pi/2) = -1$.

Solution:

Taking Laplace Transforms of both sides,

$$L[y''(t)] + 9L[y(t)] = L\{\cos 2t\}$$

$$\blacktriangleright \{s^2 L[y(t)] - sy(0) - y'(0)\} + 9L[y(t)] = \frac{s}{s^2 + 4}$$

Since $y'(0)$ is not given, we assume $y'(0) = a$

$$\blacktriangleright (s^2 + 9)L[y(t)] = s + a + \frac{s}{s^2 + 4}$$

$$\blacktriangleright L[y(t)] = \frac{s+a}{s^2+9} + \frac{s}{(s^2+9)(s^2+4)}$$

$$L[y(t)] = \frac{s}{s^2+9} + \frac{a}{s^2+9} + \frac{(\frac{1}{5})s}{(s^2+4)} + \frac{(\frac{-1}{5})s}{(s^2+9)} \text{ (by Partial Fraction Method)}$$

Take inverse Laplace transform on both sides ,

$$y(t) = L^{-1} \left(\frac{s}{s^2+9} \right) + L^{-1} \left(\frac{a}{s^2+9} \right) + \frac{1}{5} L^{-1} \left(\frac{s}{(s^2+4)} \right) - \frac{1}{5} L^{-1} \left(\frac{s}{(s^2+9)} \right)$$

$$y(t) = \frac{4}{5} \cos 3t + \frac{a}{3} \sin 3t + \frac{1}{5} \cos 2t$$

Given $y(t) = -1$ at $t = \frac{\pi}{2}$

$$y\left(\frac{\pi}{2}\right) = \frac{4}{5} \cos 3 \frac{\pi}{2} + \frac{a}{3} \sin 3 \frac{\pi}{2} + \frac{1}{5} \cos 2 \frac{\pi}{2}$$

$$-1 = \frac{-a}{3} - \frac{1}{5}$$

$$a = \frac{12}{5}$$

Therefore,

$$y(t) = \frac{4}{5} \cos 3t + \frac{4}{5} \sin 3t + \frac{1}{5} \cos 2t$$



THANK YOU

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